Environmental Kuznets Curve: Brazil (1970-2024)

1. Introduction & Variable Transformation

2. This study examines the Environmental Kuznets Curve (EKC) hypothesis for Brazil using annual data from 1970 to 2024. The aim is to investigate the relationship between economic growth and CO2 emissions per capita and to see whether environmental pollution decreases as income increases.
3. All variables are transformed into natural logarithms to make the analysis easier and examine the relationship between income and CO2 emissions.

- **LNGDPPC:** Natural logarithm of real GDP per capita (constant 2015 US$).
- **LNCO2:** Natural logarithm of CO2 emissions per capita.
- **LNGDPPC²:** Squared term of the natural logarithm of real GDP per capita, $(\ln(GDPPC))^2$.

The estimated model is: $logco2 = \beta0 + \beta1 \cdot loggdppc + \beta2 \cdot gdppc2 + \varepsilon$

An inverted-U shape consistent with the EKC requires $\beta1 > 0$ and $\beta2 < 0$. The sample contains 55 annual observations (1970-2024).

2. Scatter Plot and Regression Line

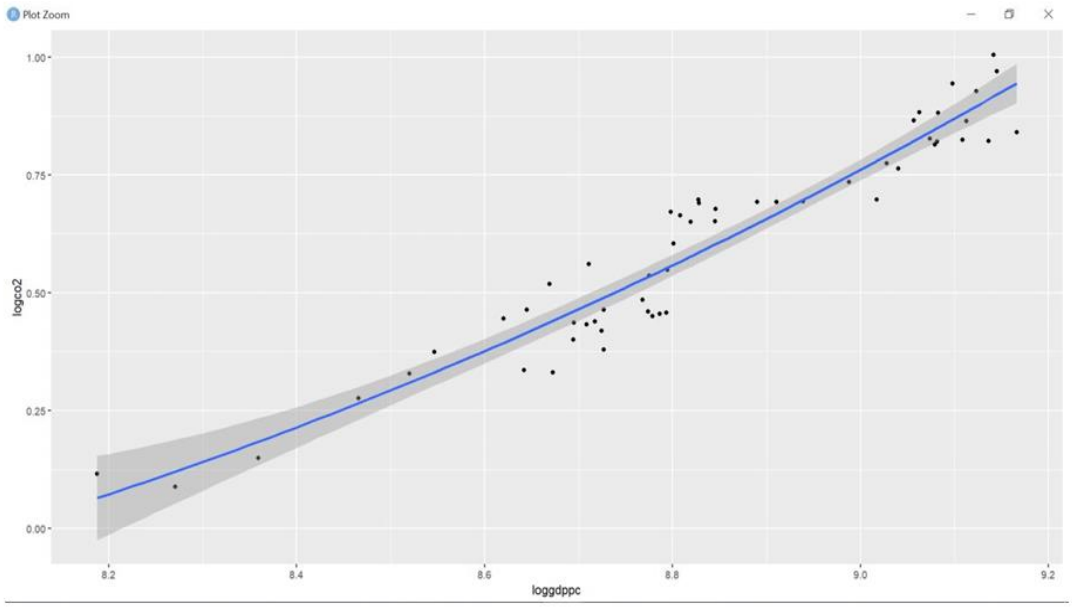


Figure 1: Scatter plot of logco2 vs. loggdppc with fitted quadratic regression line (1970-2024)

3.Model Specification & OLS Regression Results

```
> summary(fit)

Call:
lm(formula = logco2 ~ loggdppc + gdppc2, data = ekc)

Residuals:
      Min        1Q    Median        3Q       Max
-0.110315 -0.042537 -0.004049  0.049528  0.114797

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  11.6261     9.7590   1.191   0.2389
loggdppc     -3.4761     2.2270  -1.561   0.1246
gdppc2        0.2521     0.1270   1.985   0.0524 .
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.0652 on 52 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.9188,    Adjusted R-squared:  0.9157
F-statistic: 294.1 on 2 and 52 DF,  p-value: < 2.2e-16

> |
```

Table 1: OLS regression output for the quadratic EKC model (1970-2024, N=55)

4. Econometric Interpretation

I examined the relationship between CO2 emissions per capita and GDP per capita in Brazil from 1970 to 2024. The results show that the GDP coefficient is negative (-3.476), while the squared GDP coefficient is positive (0.252). This is the opposite of what we would expect if the Environmental Kuznets Curve (EKC) hypothesis was correct. Normally, we expect emissions to increase at first as income grows and then start to decrease when income reaches a certain level.

In this case, the results suggest a U-shaped relationship instead. The turning point is around 6.89 in logs, which is approximately $990. However, Brazil's lowest income level in the sample is already around $3,600. This means that Brazil is already on the increasing part of the curve, so CO2 emissions per capita continue to rise as income increases. The scatter plot also shows this upward trend.

The results are not very strong statistically either. The p-value of the linear term is 0.125, while the squared term has a p-value of 0.052. This means that the linear coefficient is not statistically significant, and the squared term is only significant at the 10% level. Although the R-squared is quite high (0.919), this may mainly be because both GDP and CO2 emissions have increased over time.

Overall, the results do not support the EKC hypothesis for Brazil during 1970–2024. There is no clear evidence that CO2 emissions start to decrease after income reaches a certain level. Instead, emissions generally continue to increase as GDP per capita grows.

Data Source and References

The data used in this study were obtained from the World Bank's World Development Indicators (WDI) database. The dataset includes annual information on GDP per capita and CO2 emissions per capita for Brazil, covering the period from 1970 to 2024.

Source: World Bank. (n.d.). *World Development Indicators*. https://databank.worldbank.org/source/world-development-indicators